

JEE MAIN REVISION MASTER: LOGARITHMS 2026

0 QUESTIONS 0% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 2 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Standard	Kcet Logarithms - Core Application	2 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Logarithms - Core Application (Standard Yield — 2 PYQs)

Core Principle: Key problem pattern in Kcet Logarithms.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which one of the following observations is correct for the features of logarithm function to any base $b > 1$?..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "The value of $e^{\log_{10} \tan 1^\circ + \log_{10} \tan 2^\circ + \log_{10} \tan 3^\circ + \dots + \log_{10} \tan 89^\circ}$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

Which one of the following observations is correct for the features of logarithm function to any base $b > 1$?

- (A) The domain of the logarithm function is $\mathbb{R}$, the set of real numbers.
- (B) The range of the logarithm function is $\mathbb{R}^+$, the set of all positive real numbers.
- (C) The point (1,0) is always on the graph of the logarithm function.
- (D) The graph of the logarithm function is decreasing as we move from left to right.

Correct Answer: None

Solution Analysis: ■ Domain $\in \mathbb{R}^+$ and range $\in \mathbb{R}$ and graph of the logarithm function is increasing. So, point (1,0) is always on the graph of logarithm function.